THE WEIL PAIRING, ENDOMORPHISMS AND AUTOMORPHISMS: SOLUTIONS

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Section 1 shows that the determinant of the mod N Galois representation is the cyclotomic character. Section 2 exhibits an endomorphism ring that is an order in a definite quaternion algebra, the third case of the classification [Sil09, Theorem III.9.3], on a single curve over $\mathbb{F}_9$. Section 3 computes the automorphism groups of order 12 and 24 in characteristic 3 and 2, the two cases of [Sil09, Theorem III.10.1] in which the group is not cyclic.

Reading. [Sil09, §§III.8–III.10], with [Sil09, Appendix A.1] for the automorphism groups in characteristic 2 and 3. For the quaternion background: [Voi21, Definition 2.2.1] for the algebra $(a, b \mid \mathbb{Q})$, [Voi21, §§3.2–3.3] for the standard involution, the reduced trace and the reduced norm, [Voi21, Main Theorem 5.4.4 and §14.1] for split and definite algebras, and [Voi21, Definitions 10.2.1 and 10.4.1] for orders and maximal orders. Proposition 2.6, Theorem 2.7 and Proposition 2.8 are the results we actually use.

`Magma`**.** If you have no local installation, every block below runs in the free online calculator, which caps how long a single job may take.

Standing notation.

- A field k, with an algebraic closure $\overline{k}$ and the separable closure $k^{\mathrm{sep}} \subseteq \overline{k}$.
- Multiplication by m, written $[m]$, and the dual $\hat{\varphi}$ of an isogeny φ.
- The q-power Frobenius endomorphism $\mathsf{F} \in \mathrm{End}(E)$ of a curve over $\mathbb{F}_q$. Beware that φ_ℓ already means the map induced by φ on $T_\ell(E)$, so the Frobenius *endomorphism* is F and never φ_q.

How to use this. Allow about 20 minutes for §1, 25 for §2 and 25 for §3. In Problem 1.5 the heart is part (3), and parts (4) and (5) are corollaries to finish at home. In Problem 2.9, parts (1)–(3) are the session and parts (4) and (5) the payoff. They can be started there and finished at home. In Problem 3.5, do characteristic 3 (parts (1)–(3)) in the session and characteristic 2 (parts (4) and (5)) at home: it is the same technique with more bookkeeping. The `Magma` parts need no time in the session: do the hand work there and run the code afterwards, at a keyboard, as a check on what you wrote. `Magma` can be slow to start, and a few of the calls below take a minute or two to return. None of them takes longer.

1. The Weil pairing and the cyclotomic character

Fix $N \geqslant 2$ with $\mathrm{char}\, k \nmid N$, so that $E[N](k^{\mathrm{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.

Definition 1.1 (The Weil pairing)**.** We write

$$e_N \colon E[N](k^{\mathrm{sep}}) \times E[N](k^{\mathrm{sep}}) \to \mu_N(k^{\mathrm{sep}})$$

for the Weil pairing.

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Definition 1.2 (Galois representations)**.** We write

$$\bar{\rho}_{E,N}\colon \mathrm{Gal}_k \to \mathrm{Aut}\big(E[N](k^{\mathrm{sep}})\big) \cong \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$$

for the mod N Galois representation, the isomorphism depending on a choice of basis, and $\rho_{E,\ell}$ for the ℓ-adic representation on $T_\ell(E)$.

Proposition 1.3 (Properties of the Weil pairing [Sil09, Propositions III.8.1 and III.8.2])**.** *The Weil pairing e_N is*

(1) bilinear;
(2) alternating: $e_N(T,T) = 1$, so $e_N(S,T) = e_N(T,S)^{-1}$;
(3) non-degenerate: if $e_N(S,T) = 1$ for all $S \in E[N](k^{\mathrm{sep}})$, then $T = O$;
(4) Galois equivariant: $\sigma\big(e_N(S,T)\big) = e_N(\sigma(S),\sigma(T))$ for $\sigma \in \mathrm{Gal}_k$;
(5) compatible: $e_{NN'}(S,T) = e_N([N']S,T)$;
(6) adjoint for isogenies: $e_N(\varphi S,T) = e_N(S,\hat{\varphi}T)$.

These properties are.

Proposition 1.4 (The determinant of φ_ℓ [Sil09, Proposition III.8.6])**.** *Let $\varphi \in \mathrm{End}(E)$ and let $\ell \neq \mathrm{char}\, k$ be a prime. Then $\det \varphi_\ell = \deg \varphi$.*

Problem 1.5 (The determinant is the cyclotomic character)**.**

(1) Let $\zeta \in \mu_N(k^{\mathrm{sep}})$ be primitive. Show that for each $\sigma \in \mathrm{Gal}_k$ there is a unique $\chi_N(\sigma) \in \mathbb{Z}/N\mathbb{Z}$ with $\sigma(\zeta) = \zeta^{\chi_N(\sigma)}$, that $\chi_N(\sigma)$ is a unit, and that the resulting map $\chi_N\colon \mathrm{Gal}_k \to (\mathbb{Z}/N\mathbb{Z})^\times$ is a homomorphism independent of the choice of ζ. This is the mod N cyclotomic character.

Solution. Since σ fixes 1 and is a ring homomorphism, it permutes the roots of X^N-1, so $\sigma(\zeta) \in \mu_N(k^{\mathrm{sep}})$. As ζ is primitive, $\mu_N(k^{\mathrm{sep}}) = \{\zeta^j : 0 \leqslant j < N\}$ and these are distinct, so $\sigma(\zeta) = \zeta^j$ for a unique $j =: \chi_N(\sigma) \bmod N$. Applying σ^{-1} shows $\zeta \mapsto \zeta^{\chi_N(\sigma)}$ is bijective on $\mu_N(k^{\mathrm{sep}})$, so $\gcd(\chi_N(\sigma), N) = 1$. For the homomorphism property, $\sigma\tau(\zeta) = \sigma(\zeta^{\chi_N(\tau)}) = \zeta^{\chi_N(\sigma)\chi_N(\tau)}$. Independence of ζ: any other primitive root is ζ^u with u a unit, and $\sigma(\zeta^u) = (\zeta^u)^{\chi_N(\sigma)}$.

(2) Let S,T be a basis of $E[N](k^{\mathrm{sep}})$ and put $\zeta \coloneqq e_N(S,T)$. Show that ζ is a *primitive* N-th root of unity, using only parts (1)–(3) of Proposition 1.3.

Solution. Let m be the order of $\zeta = e_N(S,T)$, so $m \mid N$. By bilinearity, $e_N(S,[m]T) = e_N(S,T)^m = 1$, and by (2) together with bilinearity, $e_N(T,[m]T) = e_N(T,T)^m = 1$. Since S,T generate $E[N](k^{\mathrm{sep}})$, bilinearity gives $e_N(X,[m]T) = 1$ for every $X \in E[N](k^{\mathrm{sep}})$. Non-degeneracy ((3), in the first variable, via (2)) forces $[m]T = O$. But T has order N, so $N \mid m$, and hence $m = N$.

(3) Write $\sigma(S) = aS + cT$ and $\sigma(T) = bS + dT$, so that $\bar{\rho}_{E,N}(\sigma) = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ in the basis S,T. Expand $e_N(\sigma(S),\sigma(T))$ and deduce

$$\det \bar{\rho}_{E,N} = \chi_N.$$

Solution. Using bilinearity and then $e_N(S,S) = e_N(T,T) = 1$ and $e_N(T,S) = e_N(S,T)^{-1}$,

$e_N(\sigma(S),\sigma(T)) = e_N(aS+cT,\, bS+dT) = e_N(S,S)^{ab} e_N(S,T)^{ad} e_N(T,S)^{cb} e_N(T,T)^{cd} = \zeta^{ad-bc}$.

On the other hand Galois equivariance gives $e_N(\sigma(S),\sigma(T)) = \sigma(\zeta) = \zeta^{\chi_N(\sigma)}$. So $\zeta^{ad-bc} = \zeta^{\chi_N(\sigma)}$, and since ζ has order exactly N by (2), we get $ad - bc \equiv \chi_N(\sigma) \pmod N$, i.e. $\det \bar{\rho}_{E,N}(\sigma) = \chi_N(\sigma)$.

(4) Take $k = \mathbb{Q}$. Show that $\det \bar\rho_{E,N}$ is surjective for *every* elliptic curve E over $\mathbb{Q}$, that $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}\big(E[N]\big)$, and that $\det \bar\rho_{E,N}(c) = -1$ for c a complex conjugation: the representation is odd.

Solution. Over $\mathbb{Q}$, restriction gives $\mathrm{Gal}_{\mathbb{Q}} \twoheadrightarrow \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$, and this isomorphism *is* χ_N. So χ_N, hence $\det \bar\rho_{E,N}$, is surjective, for every E. Since $\chi_N = \det \bar\rho_{E,N}$ factors through $\mathrm{Gal}\big(\mathbb{Q}(E[N])/\mathbb{Q}\big)$, its kernel contains $\mathrm{Gal}\big(\overline{\mathbb{Q}}/\mathbb{Q}(E[N])\big)$, so $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}(E[N])$. Finally $c(\zeta_N) = \zeta_N^{-1}$, so $\chi_N(c) = -1$. Taking $N = \ell^n$ and passing to the limit, $\det \rho_{E,\ell}(c) = -1$.

Remark. The determinant never degenerates, however small the image of $\bar\rho_{E,N}$ itself may be.

(5) Take $k = \mathbb{F}_q$ with q prime to N. Show that $\det \bar\rho_{E,N}(\mathsf{F}) = q$, and check this against the identity $\det \varphi_\ell = \deg \varphi$.

Solution. $\mathrm{Gal}_{\mathbb{F}_q}$ is topologically generated by the q-power Frobenius, which acts on $\mu_N(\overline{\mathbb{F}}_q)$ by $\zeta \mapsto \zeta^q$, so $\chi_N(\mathsf{F}) = q$ and (3) gives $\det \bar\rho_{E,N}(\mathsf{F}) = q$. The second computation is different: $\det \mathsf{F}_\ell = \deg \mathsf{F} = q$ because the pairing on $T_\ell(E)$ is alternating, so an endomorphism scales it by its determinant, while adjointness scales it by $\widehat{\mathsf{F}}\,\mathsf{F} = [\deg \mathsf{F}]$.

Remark. So the first computation uses Galois equivariance (4) and the second uses adjointness (6). That the two agree is the statement that $\det \rho_{E,\ell}$ is the cyclotomic character.

(6) Check (5) numerically in `Magma`, taking $N = 3$, $q = 5$ and $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$.

```
q := 5;  N := 3;
E := EllipticCurve([GF(q) | 0, 1]);      // y^2 = x^3 + 1
#E;                                      // 6
Ek := ChangeRing(E, GF(q^2));
#Ek, AbelianGroup(Ek);                   // 36, Z/6 + Z/6
T3 := [P : P in Points(Ek) | N*P eq Ek!0];
#T3;                                     // 9
S := [P : P in T3 | P ne Ek!0][1];
T := [P : P in T3 | P ne Ek!0 and P ne S and P ne -S][1];
zeta := WeilPairing(S, T, N);  Order(zeta);    // 3: a primitive cube root
Frob := func<P | P eq Ek!0 select Ek!0 else Ek![P[1]^q, P[2]^q]>;
WeilPairing(Frob(S), Frob(T), N) eq zeta^q;    // true
```

Two lines carry the mathematics. That `Order(zeta)` is 3 is (2). That the last line is `true` is $\det \bar\rho_{E,3}(\mathsf{F}) = q$, which is (5) on this curve. Note that `Frob` raises both coordinates to the q-th power: that is the q-power Frobenius, an endomorphism because E is defined over $\mathbb{F}_q$. Now write the matrix of F in the basis S, T and read its determinant off:

```
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(S) ];   // [ <1, 0> ]
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(T) ];   // [ <0, 2> ]
GF(3)!q;                                             // 2
```

Check that a different choice of basis moves the matrix but not the determinant.

Solution. $\left[\begin{smallmatrix}1 & 0\\ 0 & 2\end{smallmatrix}\right]$, **with determinant $2 = q$ mod 3.** The pairing $e_3(S,T)$ has order exactly 3, as (2) predicts, and `WeilPairing(Frob(S), Frob(T), N) eq zeta^q` is exactly $\det \bar\rho_{E,3}(\mathsf{F}) = q$. The last block gives $\mathsf{F}S = S$ and $\mathsf{F}T = [2]T$, so the matrix of F in the basis S, T is $\left[\begin{smallmatrix}1 & 0\\ 0 & 2\end{smallmatrix}\right]$, with determinant $2 = q \bmod 3$. Another basis gives another matrix, never another determinant.

Remark 1.6. Part (3) of Problem 1.5 gives back [Sil09, Corollary III.8.1.1]: if $E[N](k^{\mathrm{sep}}) \subseteq E(k)$, then $\bar{\rho}_{E,N}$ is trivial, so χ_N is trivial, so every $\sigma \in \mathrm{Gal}_k$ fixes a primitive N-th root of unity, and $\mu_N(k^{\mathrm{sep}}) \subseteq k$.

A word, for the curious, on the example in Problem 1.5(6): choosing it was half of that part. We want $q \not\equiv 1 \pmod{N}$, so that χ_N is *not* trivial and the check has content. By Remark 1.6 that already forces $E[N](k^{\mathrm{sep}}) \not\subseteq E(\mathbb{F}_q)$, so the N-torsion becomes rational only over an extension. The curve $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$ has $\#E(\mathbb{F}_5) = 6$ and $E(\mathbb{F}_{25}) \cong (\mathbb{Z}/6\mathbb{Z})^2$, so all of $E[3](\overline{\mathbb{F}}_5)$ is already rational over the quadratic extension: small enough to list, large enough to have something to say.

2. A quaternionic endomorphism ring

The classification of [Sil09, Theorem III.9.3] puts the endomorphism ring of an elliptic curve in one of three cases: $\mathbb{Z}$, an order in an imaginary quadratic field, or an order in a definite quaternion algebra over $\mathbb{Q}$. The first two are easy to exhibit: a generic curve has endomorphism ring $\mathbb{Z}$, and $y^2 = x^3 + x$ has $\mathbb{Z}[i]$ inside its endomorphism ring. Here is an example of the third, on the smallest curve that has one.

Definition 2.1 (The quaternion algebra D [Voi21, Definition 2.2.1])**.** Let $D \coloneqq (-1, -3 \mid \mathbb{Q})$, the quaternion algebra with $i^2 = -1$, $j^2 = -3$ and $ij = -ji$. Its reduced discriminant is the product of the primes p at which $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is not split.

Definition 2.2 (Definite quaternion algebras [Voi21, Main Theorem 5.4.4 and §14.1])**.** A quaternion algebra B over $\mathbb{Q}$ is definite when $B \otimes_{\mathbb{Q}} \mathbb{R}$ is Hamilton's $\mathbb{H}$ rather than $M_2(\mathbb{R})$, equivalently when its reduced norm is a positive definite quadratic form in the four coordinates. Otherwise B is indefinite.

Definition 2.3 (Orders and the reduced discriminant [Voi21, §15.2])**.** Orders are as in [Voi21, Definition 10.2.1]. The reduced discriminant $\mathrm{discrd}(\mathcal{O})$ of an order $\mathcal{O} \subseteq D$ is the positive integer with $\mathrm{discrd}(\mathcal{O})^2 = \left|\det\big(\mathrm{trd}(\alpha_k \alpha_l)\big)_{k,l}\right|$ for a $\mathbb{Z}$-basis $\alpha_1, \dots, \alpha_4$ of $\mathcal{O}$.

Proposition 2.4 (The characteristic polynomial of Frobenius [Sil09, §V.2])**.** *Let E be an elliptic curve over $\mathbb{F}_q$ with F its q-power Frobenius, $a \coloneqq q + 1 - \#E(\mathbb{F}_q)$, and $\ell \nmid q$ a prime. Then F satisfies $\mathsf{F}^2 - [a]\mathsf{F} + [q] = 0$ in $\mathrm{End}(E)$: by [Sil09, Proposition III.8.6] the characteristic polynomial of F_ℓ is $T^2 - aT + q$, and every $\varphi \in \mathrm{End}(E)$ satisfies $\varphi^2 - [\mathrm{tr}\,\varphi_\ell]\varphi + [\deg \varphi] = 0$ in $\mathrm{End}(E)$, not merely on $T_\ell(E)$.*

Corollary 2.5 (Factoring through a separable isogeny [Sil09, Corollary III.4.11])**.** *If $\varphi\colon E_1 \to E_2$ is separable and $\psi\colon E_1 \to E_3$ is an isogeny with $\ker\varphi \subseteq \ker\psi$, then $\psi = \lambda \circ \varphi$ for a unique isogeny $\lambda\colon E_2 \to E_3$.*

Proposition 2.6 (Where D is split [Voi21, Exercise 11.11 and §14.1])**.** *The algebra $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is split for every prime $p \neq 3$, and $D \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is not. So D is a division algebra, and its reduced discriminant is 3.*

Theorem 2.7 (Maximal orders [Voi21, Theorem 15.5.5])**.** *An order $\mathcal{O} \subseteq D$ is maximal if and only if $\mathrm{discrd}(\mathcal{O}) = 3$.*

Proposition 2.8 (The index and the discriminant [Voi21, §15.2])**.** *If $\mathcal{O} \subseteq \mathcal{O}'$ are orders of D, then $\mathrm{discrd}(\mathcal{O}) = [\mathcal{O}' : \mathcal{O}]\,\mathrm{discrd}(\mathcal{O}')$.*

For the rest of this section

$$E\colon y^2 = x^3 - x \quad \text{over } \mathbb{F}_3,$$

so that the generic E of Proposition 2.4 is this curve, with $q = 3$.

Problem 2.9 (End($E_{\mathbb{F}_9}$) is a maximal order in a quaternion algebra)**.**
Every part below has a hand computation and a check to run in `Magma`. The curve is small enough that $[i]$ and F can be verified on all sixteen points of $E(\mathbb{F}_9)$.

(1) Check that E is an elliptic curve over $\mathbb{F}_3$, list $E(\mathbb{F}_3)$, and deduce that $a = 0$ and that

$$\mathsf{F}^2 = [-3].$$

Record also that every point of $E[2](\overline{\mathbb{F}}_3)$ lies in $E(\mathbb{F}_3)$: this is what makes part (5) work.

```
E3 := EllipticCurve([GF(3) | -1, 0]);     // prints as y^2 = x^3 + 2*x
Points(E3);
#E3;                                      // 4
3 + 1 - #E3;                              // 0        (so a = 0)
jInvariant(E3);                           // 0
```

`Magma` hands you the point count and the j-invariant. The identity $\mathsf{F}^2 = [-3]$ is yours to deduce, from Proposition 2.4. Read $E[2](\overline{\mathbb{F}}_3) \subseteq E(\mathbb{F}_3)$ off the list of rational points — which points of $y^2 = f(x)$ have order 2?

Solution. $\boldsymbol{E(\mathbb{F}_3) = \{O, (0,0), (1,0), (2,0)\}}$**, so** $\boldsymbol{a = 0}$**.** Here $A = -1$ and $B = 0$, so $\Delta = -16(4A^3 + 27B^2) = 64 \equiv 1 \pmod 3$, which is non-zero: E is an elliptic curve. Every $x \in \mathbb{F}_3$ satisfies $x^3 = x$, so $x^3 - x = 0$ for all three values, giving

$$E(\mathbb{F}_3) = \{O, (0,0), (1,0), (2,0)\}, \qquad \#E(\mathbb{F}_3) = 4.$$

Hence $a = 3 + 1 - 4 = 0$, and Proposition 2.4 gives $\mathsf{F}^2 + [3] = 0$, i.e. $\mathsf{F}^2 = [-3]$. Finally the three affine rational points listed above are precisely the points of order 2, since a point (x, y) of $E(\overline{\mathbb{F}}_3)$ has order 2 if and only if $y = 0$. All three are in $E(\mathbb{F}_3)$, so $E[2](\overline{\mathbb{F}}_3) \subseteq E(\mathbb{F}_3)$, which is what `Magma`'s list of rational points shows.

(2) Show that $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so that $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. Show that

$$[i](x, y) \coloneqq (-x, iy)$$

defines an endomorphism of $E_{\mathbb{F}_9}$, and that $[i]^2 = [-1]$. Why is a morphism $E \to E$ fixing O automatically a group homomorphism? Mind Warning 2.10 before you start typing.

```
F9 := GF(9);  i := Sqrt(F9!(-1));
i^2, MinimalPolynomial(i);                // 2 ($.1^2 + 1), and 2 = -1
E9 := ChangeRing(E3, F9);  #E9;           // 16
imap := func<P | P eq E9!0 select E9!0 else E9![-P[1], i*P[2]]>;
forall{P : P in Points(E9) | imap(P) in Points(E9)};        // [i] is well defined
forall{<P,Q> : P, Q in Points(E9) | imap(P+Q) eq imap(P)+imap(Q)};  // additive
forall{P : P in Points(E9) | imap(imap(P)) eq -P};          // [i]^2 = [-1]
// all true
```

The second `forall` is additivity, checked on all 16^2 pairs. That is a proof only because the curve is finite, and it is no answer to the question above: additivity is not a coincidence to be confirmed point by point but a consequence of fixing O.

Solution. $x^2 + 1$ has no root in $\mathbb{F}_3$: the values of $x^2 + 1$ are $1, 2, 2$. So it is irreducible and $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. For the map, substitute:

$$(iy)^2 = i^2y^2 = -(x^3 - x) = -x^3 + x = (-x)^3 - (-x).$$

So $[i]$ carries points of E to points of E, and it is given by polynomials in x, y with coefficients in $\mathbb{F}_9$, hence is a morphism $E_{\mathbb{F}_9} \to E_{\mathbb{F}_9}$. It fixes $O = (0:1:0)$. A morphism $\varphi\colon E_{\mathbb{F}_9} \to E_{\mathbb{F}_9}$ with $\varphi(O) = O$ is a group homomorphism because the group law of E is the one of $\mathrm{Pic}^0 E$, transported along the isomorphism of groups $\kappa\colon E(\overline{\mathbb{F}}_3) \to \mathrm{Pic}^0 E_{\overline{\mathbb{F}}_3}$, $P \mapsto [(P) - (O)]$. If φ is constant it is the zero map. Otherwise it is finite, so the push-forward of divisors φ_* is a homomorphism $\mathrm{Pic}^0 E_{\overline{\mathbb{F}}_3} \to \mathrm{Pic}^0 E_{\overline{\mathbb{F}}_3}$, and since $\varphi(O) = O$ it sends $[(P) - (O)]$ to $[(\varphi(P)) - (O)]$ for every point P of $E(\overline{\mathbb{F}}_3)$. Hence $\varphi = \kappa^{-1} \circ \varphi_* \circ \kappa$ is a composition of homomorphisms [Sil09, Theorem III.4.8]. So $[i] \in \mathrm{End}(E_{\mathbb{F}_9})$. Composing,

$$[i]^2(x, y) = [i](-x, iy) = (x, i^2 y) = (x, -y) = [-1](x, y),$$

so $[i]^2 = [-1]$. In particular $[i]$ is an automorphism, of order 4.

(3) Show that $\mathsf{F}[i] = -[i]\mathsf{F}$. Conclude that $\mathrm{End}(E_{\mathbb{F}_9})$ is not commutative, so that $E_{\mathbb{F}_9}$ falls into the third case of the classification.

```
Frob := func<P | P eq E9!0 select E9!0 else E9![P[1]^3, P[2]^3]>;
forall{P : P in Points(E9) | Frob(Frob(P)) eq -3*P};            // Frob^2 = [-3]
forall{P : P in Points(E9) | Frob(imap(P)) eq -imap(Frob(P))};   // Frob[i] = -[i]Frob
// both true
```

The first line verifies the $\mathsf{F}^2 = [-3]$ you deduced in (1) from the characteristic polynomial, which is worth having independently. For the second, do the computation by hand as well: it comes down to a single fact about i^3.

Solution. $\mathsf{F}(x, y) = (x^3, y^3)$, so

$$[i]\mathsf{F}(x, y) = [i](x^3, y^3) = (-x^3, iy^3),$$
$$\mathsf{F}[i](x, y) = \mathsf{F}(-x, iy) = \big((-x)^3, (iy)^3\big) = (-x^3, i^3 y^3).$$

Now $i^3 = i^2 \cdot i = -i$, so $\mathsf{F}[i](x, y) = (-x^3, -iy^3) = -[i]\mathsf{F}(x, y)$. This shows that

$$\mathsf{F}[i] = -[i]\mathsf{F}.$$

Since $[2][i]\mathsf{F} \neq 0$ (it has degree 12), $-[i]\mathsf{F} \neq [i]\mathsf{F}$: the two endomorphisms $[i]$ and F do not commute, so $\mathrm{End}(E_{\mathbb{F}_9})$ is non-commutative. By the classification it is therefore an order in a definite quaternion algebra: cases $\mathbb{Z}$ and an imaginary quadratic order are both commutative.

(4) Deduce that

$$\mathrm{End}(E_{\mathbb{F}_9}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong D = (-1, -3 \mid \mathbb{Q}).$$

Write down the reduced norm of D and check that D is definite. How does the reduced norm compare with the degree of an endomorphism?

```
D<ii,jj> := QuaternionAlgebra<Rationals() | -1, -3>;
RamifiedPrimes(D), Discriminant(D);      // [ 3 ] 3
IsDefinite(D);                           // true
Trace(jj), Norm(jj);                     // 0 3
```

The two arguments are the squares of the generators, so the algebra `Magma` builds is the D of Definition 2.1. Here `Trace` and `Norm` on a quaternion algebra are the reduced trace and the reduced norm, and the pair they return for `jj` should be recognizable as $(\mathrm{tr}\,\mathsf{F}, \deg \mathsf{F})$. All of this is about D alone. That D is the *whole* of $\mathrm{End}(E_{\mathbb{F}_9}) \otimes \mathbb{Q}$ is the one claim here `Magma` cannot be asked, and it needs a rank count.

Solution. **$\mathrm{nrd}(t+xi+yj+zij) = t^2+x^2+3y^2+3z^2$. So nrd = deg.** By (2), (1) and (3) the elements $[i]$ and F of $\mathrm{End}(E_{\mathbb{F}_9})\otimes\mathbb{Q}$ satisfy $i^2=-1$, $\mathsf{F}^2=-3$ and $i\mathsf{F}=-\mathsf{F}i$, which are the defining relations of $D=(-1,-3\mid\mathbb{Q})$. So $i\mapsto[i]$, $j\mapsto\mathsf{F}$ defines a ring homomorphism $D\to\mathrm{End}(E_{\mathbb{F}_9})\otimes\mathbb{Q}$. It is injective because D is a division algebra (Proposition 2.6), so its only two-sided ideals are 0 and D. And it is surjective because $\mathrm{End}(E_{\mathbb{F}_9})$ has rank at most 4 over $\mathbb{Z}$ ([Sil09, Corollary III.7.5]) while D has dimension exactly 4. The reduced norm of D is

$$\mathrm{nrd}(t+xi+yj+zij) = t^2+x^2+3y^2+3z^2,$$

which is positive definite: $(a,b\mid\mathbb{Q})$ is definite exactly when $a<0$ and $b<0$ [Voi21, Main Theorem 5.4.4 and §14.1], and here $a=-1$, $b=-3$. Under the isomorphism, nrd is the degree: the dual satisfies

$$\hat\varphi\varphi=[\deg\varphi],\qquad \varphi+\hat\varphi=[1+\deg\varphi-\deg(1-\varphi)]\in\mathbb{Z}$$

[Sil09, Theorem III.6.2 and Corollary III.6.3], so $\varphi\mapsto\hat\varphi$ is the standard involution of D and $\mathrm{nrd}(\varphi)=\hat\varphi\varphi=\deg\varphi$. For instance $\deg\mathsf{F}=3=\mathrm{nrd}(j)$ and $\deg[i]=1=\mathrm{nrd}(i)$.
In the code, `Trace(jj), Norm(jj)` returns 0 and 3: that is $\mathrm{trd}(\mathsf{F})=0=a$ and $\mathrm{nrd}(\mathsf{F})=3=\deg\mathsf{F}$, as the identification of nrd with the degree requires.

Remark. Note that the prime at which D is not split, by Proposition 2.6, is the characteristic: in general, when the endomorphism algebra of an elliptic curve in characteristic p is a quaternion algebra, it is the one over $\mathbb{Q}$ ramified exactly at p and ∞ [Sil09, Theorem V.3.1], [Voi21, Theorem 42.1.9].

(5) Show that $\frac{1+\mathsf{F}}{2}$ and $\frac{i+i\mathsf{F}}{2}$ lie in $\mathrm{End}(E_{\mathbb{F}_9})$. Then show that

$$\mathcal{O}\coloneqq\mathbb{Z}+\mathbb{Z}i+\mathbb{Z}\frac{1+\mathsf{F}}{2}+\mathbb{Z}\frac{i+i\mathsf{F}}{2}$$

is an order of D, compute $\mathrm{discrd}(\mathcal{O})$, and conclude with Theorem 2.7 that $\mathrm{End}(E_{\mathbb{F}_9})=\mathcal{O}$ is a maximal order.

```
OO := QuaternionOrder([D!1, ii, jj, ii*jj]);
Discriminant(OO);                          // 12
O  := QuaternionOrder([D!1, ii, (1+jj)/2, (ii+ii*jj)/2]);
Discriminant(O), IsMaximal(O);             // 3 true
```

`QuaternionOrder` takes a $\mathbb{Z}$-basis and `Discriminant` of an order is its reduced discriminant, so the two lists above are the $\mathbb{Z}$-bases of $\mathcal{O}_0$ and $\mathcal{O}$. Do not reach for `Index` to compare the two orders: Magma has no index for quaternion orders, and answers `Runtime error in 'Index': Bad argument types`. Recover $[\mathcal{O}:\mathcal{O}_0]$ from the two discriminants instead, by Proposition 2.8, and check it against the determinant of the change of basis.

The first sentence of this part is where the real work is, and none of the code touches it: $\frac{1+\mathsf{F}}{2}$ is a quotient in a ring with no division, and only the kernel argument puts it in $\mathrm{End}(E_{\mathbb{F}_9})$.

Solution. **$\mathrm{discrd}(\mathcal{O})=3$, so $\mathrm{End}(E_{\mathbb{F}_9})=\mathcal{O}$.** By (1) every $P\in E[2](\overline{\mathbb{F}}_3)$ lies in $E(\mathbb{F}_3)$, so F fixes it, and therefore

$$(1+\mathsf{F})P=P+\mathsf{F}P=P+P=[2]P=O.$$

Hence $\ker[2]=E[2]\subseteq\ker(1+\mathsf{F})$. Since the characteristic 3 does not divide 2, the isogeny $[2]$ is separable, so Corollary 2.5 gives an isogeny λ with $1+\mathsf{F}=\lambda\circ[2]$. That is, $\frac{1+\mathsf{F}}{2}=\lambda$ lies in $\mathrm{End}(E_{\mathbb{F}_9})$. Multiplying by $[i]\in\mathrm{End}(E_{\mathbb{F}_9})$ shows $\frac{i+i\mathsf{F}}{2}=[i]\circ\lambda$ lies there too.

So $\mathcal{O} \subseteq \mathrm{End}(E_{\mathbb{F}_9})$, and $\mathcal{O}$ is an order: it is a free $\mathbb{Z}$-module of rank 4 spanning D over $\mathbb{Q}$, and it is closed under multiplication, since the pairwise products of 1, i, $\frac{1+\mathsf{F}}{2}$ and $\frac{i+i\mathsf{F}}{2} = i\,\frac{1+\mathsf{F}}{2}$ lie in their $\mathbb{Z}$-span: it suffices to check the products of i and $\frac{1+\mathsf{F}}{2}$, and

$$i^2 = -1, \qquad \Big(\frac{1+\mathsf{F}}{2}\Big)^2 = \frac{1+2\mathsf{F}+\mathsf{F}^2}{4} = \frac{-2+2\mathsf{F}}{4} = \frac{1+\mathsf{F}}{2} - 1,$$
$$i\,\frac{1+\mathsf{F}}{2} = \frac{i+i\mathsf{F}}{2}, \qquad \frac{1+\mathsf{F}}{2}\,i = \frac{i-i\mathsf{F}}{2} = i - \frac{i+i\mathsf{F}}{2}.$$

To see that $\mathcal{O}$ is maximal, compute reduced discriminants, starting with the obvious suborder $\mathcal{O}_0 \coloneqq \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\mathsf{F} + \mathbb{Z}i\mathsf{F}$. Its reduced trace form $\big(\mathrm{trd}(\alpha_k\alpha_l)\big)$ in the basis $1, i, \mathsf{F}, i\mathsf{F}$ is diagonal, because distinct basis products have reduced trace 0:

$$\mathrm{trd}(1) = 2, \quad \mathrm{trd}(i^2) = -2, \quad \mathrm{trd}(\mathsf{F}^2) = -6, \quad \mathrm{trd}\big((i\mathsf{F})^2\big) = -6,$$

so the Gram determinant is $2 \cdot (-2) \cdot (-6) \cdot (-6) = -144$ and $\mathrm{discrd}(\mathcal{O}_0) = 12$. Expressing the basis of $\mathcal{O}$ in terms of that of $\mathcal{O}_0$ gives the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}, \qquad \det = \tfrac{1}{4},$$

so $[\mathcal{O} : \mathcal{O}_0] = 4$, and by Proposition 2.8

$$\mathrm{discrd}(\mathcal{O}) = \frac{\mathrm{discrd}(\mathcal{O}_0)}{[\mathcal{O} : \mathcal{O}_0]} = \frac{12}{4} = 3.$$

By Theorem 2.7, $\mathcal{O}$ is therefore a maximal order. A maximal order is not properly contained in any order, and $\mathcal{O} \subseteq \mathrm{End}(E_{\mathbb{F}_9})$ is an order, so $\mathcal{O} = \mathrm{End}(E_{\mathbb{F}_9})$.

`Magma` returns 12 and 3 for the two reduced discriminants, and the ratio $12/3 = 4$ is the index $[\mathcal{O} : \mathcal{O}_0]$ computed above from the change of basis, so the two routes agree.

Remark. Note that `IsMaximal` is not telling us anything the discriminant did not: by Theorem 2.7 maximality *is* the statement discrd = 3.

A closing word on scope: when the endomorphism ring of a curve over $\overline{\mathbb{F}}_p$ is non-commutative, all of its endomorphisms are already defined over $\mathbb{F}_{p^2}$, so this also computes $\mathrm{End}(E_{\overline{\mathbb{F}}_3}) = \mathcal{O}$, in agreement with Deuring's theorem that such a ring is a maximal order in the quaternion algebra ramified exactly at p and ∞ [Voi21, Theorem 42.1.9].

Warning 2.10 ($\mathbb{F}_9$ does not come with i)**.** `GF(9)` does *not* come with i. `Magma`'s default generator of $\mathbb{F}_9$ is a root of the Conway polynomial $x^2 + 2x + 2$, so writing `F9<i> := GF(9)` gives an i with $i^2 = i + 1$, not $i^2 = -1$, and every check in Problem 2.9 would silently fail. Construct the square root of -1 explicitly, and confirm it: `MinimalPolynomial` of the element you build must be $x^2 + 1$, and `i^2` must print as 2, which is -1 in $\mathbb{F}_9$.

3. Automorphisms in characteristic 2 and 3

For $\mathrm{char}\, k = 2, 3$ the geometric automorphism group of an elliptic curve may be non-abelian, which is the subject of Problem 3.5. In the following statements, E is an elliptic curve over k, given by a Weierstrass equation with coefficients a_1, a_2, a_3, a_4, a_6.

Definition 3.1 (Changes of variables)**.** A change of variables preserving a Weierstrass equation is a substitution

$$x = u^2x' + r, \qquad y = u^3y' + u^2sx' + t, \qquad u \in \overline{k}^\times,\ r, s, t \in \overline{k}. \tag{3.1}$$

For such u, r, s, t we write

$$\Psi_{u,r,s,t}(x, y) := (u^2 x + r,\ u^3 y + u^2 s x + t).$$

Theorem 3.2 (The automorphism group [Sil09, Theorem III.10.1])**.** *The group* $\mathrm{Aut}(E_{\overline{k}})$ *is finite, of order*

$$\#\,\mathrm{Aut}(E_{\overline{k}}) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \operatorname{char} k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \operatorname{char} k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 2. \end{cases}$$

Corollary 3.3 (The cyclic cases [Sil09, Corollary III.10.2])**.** *If* $\operatorname{char} k \neq 2, 3$*, then* $\mathrm{Aut}(E_{k^{\mathrm{sep}}}) \cong \mu_n(k^{\mathrm{sep}})$ *with* $n = \#\,\mathrm{Aut}(E_{\overline{k}})$*, via* $[\zeta](x, y) := (\zeta^2 x, \zeta^3 y)$*. In particular, the geometric automorphism group is cyclic in that case.*

Proposition 3.4 (The action on the coefficients [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Under a change of variables* (3.1) *the coefficients transform by*

$$\begin{aligned} u a_1' &= a_1 + 2s, \\ u^2 a_2' &= a_2 - s a_1 + 3r - s^2, \\ u^3 a_3' &= a_3 + r a_1 + 2t, \\ u^4 a_4' &= a_4 - s a_3 + 2r a_2 - (t + rs) a_1 + 3r^2 - 2st, \\ u^6 a_6' &= a_6 + r a_4 + r^2 a_2 + r^3 - t a_3 - t^2 - rt a_1. \end{aligned} \tag{3.2}$$

A substitution (3.1) *carries* E *to the curve with coefficients* a_i'*, so it is a change of coordinates from* E *to itself exactly when* $a_i' = a_i$ *for every* i*, and then* $\Psi_{u,r,s,t}$ *is the automorphism of* E *that it defines. Setting* $a_i' = a_i$ *in* (3.2) *therefore turns those five lines into five equations in the four unknowns* u, r, s, t*, whose solutions are exactly the elements of* $\mathrm{Aut}(E_{\overline{k}})$*.*

The whole point of this section is visible in (3.2): observe the 2's and 3's in the coefficients. The characteristic enters through the terms $2s$, $2t$ and $3r$. Away from 2 and 3 they are invertible multiples of s, t and r, which determines those three and leaves u as the only free parameter: that is why the group is cyclic there. In characteristic 3 the term $3r$ vanishes, in characteristic 2 the terms $2s$ and $2t$ do, and the parameters they no longer pin down are what make the two groups of Problem 3.5 large, and non-cyclic.

Problem 3.5 (The two non-cyclic cases)**.**
As in Problem 2.9, each part pairs a hand computation with a check in `Magma`. Here the machine is best at naming the group, which is the trickiest thing to do by hand.

(1) Let $\operatorname{char} k = 3$. Show that every short Weierstrass equation $y^2 = x^3 + Ax + B$ has $j = 0$, and explain why a curve with $j \neq 0$ in characteristic 3 needs an $a_2 x^2$ term. So the case "$j = 0 = 1728$, $\operatorname{char} k = 3$" of Theorem 3.2 is exactly the short-Weierstrass one. The first claim is a one-line sweep over the whole family:

```
{ jInvariant(EllipticCurve([GF(3) | A, B])) : A, B in GF(3) | A ne 0 };  // { 0 }
jInvariant(EllipticCurve([GF(3) | 0, 1, 0, 0, -1]));   // 1
```

The condition `A ne 0` is what keeps the equation non-singular: work out what $\Delta \neq 0$ becomes for $y^2 = x^3 + Ax + B$ once $27 = 0$, and you will find that B drops out. The set that comes back

has one element, which is the first claim. The second line is a curve with an a_2x^2 term, and its j-invariant is not in that set.

SOLUTION. For $y^2 = x^3 + Ax + B$ we have $c_4 = -48A$ and $c_6 = -864B$. In characteristic 3, $48 = 3 \cdot 16 \equiv 0$, so $c_4 = 0$, and therefore $j = c_4^3/\Delta = 0$. (Also $1728 = 3^3 \cdot 64 \equiv 0$, so $j = 0 = 1728$: the two exceptional values of Theorem 3.2 collide.) A curve with $j \neq 0$ in characteristic 3 therefore cannot be put in short Weierstrass form. Indeed, completing the cube is what removes a_2, and it requires dividing by 3. In characteristic 3 one can only complete the square, reaching $y^2 = x^3 + a_2x^2 + a_4x + a_6$, and $j \neq 0$ forces $a_2 \neq 0$. So the case at hand is the short-Weierstrass one, which is where $E\colon y^2 = x^3 - x$ of Problem 2.9 lives: Problem 2.9(1) found $j = 0$ for it.
As for the sweep: in characteristic 3 we have $27 = 0$, so

$$\Delta = -16(4A^3 + 27B^2) = -64A^3 = 2A^3,$$

and non-singularity is just $A \neq 0$ — B is unconstrained, which is why it is not mentioned in the condition. The one-element set that comes back says that every short Weierstrass curve over $\mathbb{F}_3$ has $j = 0$, while $y^2 = x^3 + x^2 - 1$, which has an a_2 term, has $j = 1 \neq 0$.

(2) Let $\operatorname{char} k = 3$ and $E\colon y^2 = x^3 - x$. Using (3.2), show first that $s = t = 0$, and then that (3.1) is an automorphism of E if and only if

$$u^4 = 1 \qquad \text{and} \qquad r^3 - r = 0,$$

i.e. $u \in \mu_4(\overline{\mathbb{F}}_3) \subseteq \mathbb{F}_9^\times$ and $r \in \mathbb{F}_3$. Conclude $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) = 12$, and that all 12 are already defined over $\mathbb{F}_9$. Where did r become free, and what would have forced $r = 0$ if $\operatorname{char} k \neq 3$? Two counts settle both claims at once:

```
E3 := EllipticCurve([GF(3) | -1, 0]);
#AutomorphismGroup(E3), #AutomorphismGroup(ChangeRing(E3, GF(9)));   // 6 12
```

Predict both numbers from your conditions on u and r before running it. Over $\mathbb{F}_3$ the fourth roots of unity are only ± 1, while r already ranges over all of $\mathbb{F}_3$. Over $\mathbb{F}_9$ all four roots of unity appear. The second number being 12, and being reached over $\mathbb{F}_9$ rather than over some larger field, is exactly the claim that all twelve automorphisms are defined there.

SOLUTION. **$\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) = 12$, all defined over $\mathbb{F}_9$. Over $\mathbb{F}_3$ there are 6.** Here $a_1 = a_2 = a_3 = 0$, $a_4 = A = -1$, $a_6 = B = 0$. From (3.2), $ua_1' = 2s$ and $u^3a_3' = 2t$. Since 2 is invertible in characteristic 3, requiring $a_1' = a_3' = 0$ gives $s = t = 0$. With $s = t = 0$ the remaining equations read

$$u^2a_2' = 3r = 0, \qquad u^4a_4' = A + 3r^2 = A, \qquad u^6a_6' = B + rA + r^3.$$

The first is automatic—this is where r becomes free, because $3r = 0$ in characteristic 3. If $\operatorname{char} k \neq 2, 3$, the same equation would read $3r = 0$ with 3 invertible, forcing $r = 0$, and only u would survive: that is exactly why $\operatorname{Aut}(E_{\overline{k}}) \cong \mu_n(\overline{k})$ away from 2 and 3. Requiring $a_4' = A$ gives $A/u^4 = A$, so $u^4 = 1$ as $A \neq 0$. Requiring $a_6' = B$ gives $r^3 + Ar + B = u^6B$, which for $A = -1$, $B = 0$ is $r^3 - r = 0$, i.e. $r \in \mathbb{F}_3$. So there are 4 choices of u (namely $\mu_4(\overline{\mathbb{F}}_3) = \{\pm 1, \pm i\} \subseteq \mathbb{F}_9^\times$, using i from Problem 2.9(2)) and 3 of r: twelve in all, and since $\mu_4(\overline{\mathbb{F}}_3) \cup \mathbb{F}_3 \subseteq \mathbb{F}_9$, all twelve are defined over $\mathbb{F}_9$.
The prediction for the two counts: over $\mathbb{F}_3$ only $u = \pm 1$ satisfy $u^4 = 1$, while r still ranges over $\mathbb{F}_3$, giving $2 \cdot 3 = 6$; over $\mathbb{F}_9$ all of $\mu_4(\overline{\mathbb{F}}_3)$ is available, giving $4 \cdot 3 = 12$. `Magma` returns `6 12`. The

12 is $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3})$ by Theorem 3.2, so passing to any field larger than $\mathbb{F}_9$ cannot produce more: that is the precise sense in which all twelve are already defined over $\mathbb{F}_9$.

(3) Write $\varphi_{u,r} := \Psi_{u,r,0,0}$, so that $\varphi_{u,r}(x,y) = (u^2x + r, u^3y)$. Compute $\varphi_{u',r'} \circ \varphi_{u,r}$ and deduce

$$\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z},$$

where $\mathbb{Z}/3\mathbb{Z} = \mathbb{F}_3$ is the group of translations r, where $\mathbb{Z}/4\mathbb{Z} = \mu_4(\overline{\mathbb{F}}_3)$ is the group of the u, and where u acts on r by $r \mapsto u^2 r$. Since $u^4 = 1$ forces $u^2 = \pm 1$, that action is not faithful: it factors through the squaring map $\mu_4(\overline{\mathbb{F}}_3) \twoheadrightarrow \{\pm 1\} = \operatorname{Aut}(\mathbb{Z}/3\mathbb{Z})$, so ± 1 act trivially and $\pm i$ act by $r \mapsto -r$. Show this group is non-abelian with a *unique* element of order 2, and deduce that it is neither cyclic, nor dihedral, nor A_4: it is the dicyclic group of order 12. Which automorphism is the unique involution? `AutomorphismGroup` returns a polycyclic group and `IdentifyGroup` gives its label in the small-groups database.

```
G3 := AutomorphismGroup(ChangeRing(E3, GF(9)));
IdentifyGroup(G3), GroupName(G3);        // <12, 1>  C3:C4
IsAbelian(G3), IsCyclic(G3);             // false false
{* Order(g) : g in G3 *};                // {* 1, 2, 3^^2, 4^^6, 6^^2 *}
```

The last line is the multiset of element orders. Say how it rules out the dihedral group and A_4 — how many involutions does each of those have? — and match its single element of order 2 against the automorphism you named.

Solution. **The dicyclic group $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. Its unique involution is $[-1]$.** Directly,

$$\varphi_{u',r'}\big(\varphi_{u,r}(x,y)\big) = \varphi_{u',r'}(u^2x + r,\, u^3y) = \big(u'^2u^2x + u'^2r + r',\, u'^3u^3y\big) = \varphi_{u'u,\, u'^2r+r'}(x,y),$$

so the composition law is $(u',r') \cdot (u,r) = (u'u,\, u'^2r + r')$. This is the semidirect product $\mathbb{F}_3 \rtimes \mu_4(\overline{\mathbb{F}}_3) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$, where $\mu_4(\overline{\mathbb{F}}_3)$ acts on $r \in \mathbb{F}_3$ by multiplication by u^2. Since $u^4 = 1$ we have $u^2 = \pm 1$, so the action is through the quotient $\mu_4(\overline{\mathbb{F}}_3) \to \{\pm 1\} = \operatorname{Aut}(\mathbb{Z}/3\mathbb{Z})$. For the involutions: $(u,r)^2 = (u^2,\, u^2r + r)$, which is the identity if and only if $u^2 = 1$ and $(u^2+1)r = 2r = 0$, hence $r = 0$ because 2 is invertible. So the elements of order dividing 2 are $(\pm 1, 0)$, and the unique involution is $\varphi_{-1,0}(x,y) = (x,-y) = [-1]$. The group is not cyclic, being non-abelian ($\mu_4(\overline{\mathbb{F}}_3)$ acts non-trivially on $\mathbb{F}_3$). Its unique involution rules out D_6, which has seven, and A_4, which has three. The remaining group of order 12 is the dicyclic group $\mathrm{Dic}_3 = \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. Magma's label `<12, 1>` is that group, and its multiset of element orders shows a single element of order 2, against seven in D_6 and three in A_4: the counting argument above, in one line.

(4) Let $\operatorname{char} k = 2$ and $E\colon y^2 + y = x^3$. Using (3.2), show that $a_1' = a_1$ automatically, that $a_2' = 0$ forces $r = s^2$, and then that (3.1) is an automorphism if and only if

$$u^3 = 1, \qquad s^4 = s, \qquad t^2 + t = s^6.$$

Deduce that $s \in \mathbb{F}_4$, count the solutions, and conclude $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$, all defined over $\mathbb{F}_4$. The same two counts as in (2):

```
E2 := EllipticCurve([GF(2) | 0, 0, 1, 0, 0]);          // y^2 + y = x^3
#AutomorphismGroup(E2), #AutomorphismGroup(ChangeRing(E2, GF(4)));   // 2 24
```

Predict both first. Over $\mathbb{F}_2$ the three conditions bite hard: $u^3 = 1$ leaves only $u = 1$, and of the values $s \in \mathbb{F}_2$ one of them makes $t^2 + t = s^6$ unsolvable in $\mathbb{F}_2$. The gap between the two

numbers here is much wider than in characteristic 3, and that gap is the content of "all defined over $\mathbb{F}_4$".

SOLUTION. **$\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$, all defined over $\mathbb{F}_4$. Over $\mathbb{F}_2$ there are 2.** Here $a_1 = a_2 = 0$, $a_3 = 1$, $a_4 = a_6 = 0$. From (3.2), $ua_1' = a_1 + 2s = 0$ automatically, because $2 = 0$: this is where s becomes free, the characteristic-2 counterpart of the free r in part (2). Next $u^2 a_2' = 3r - s^2 = r - s^2$ using $3 = 1$, so $a_2' = 0$ forces $r = s^2$ — which is why the substitution is usually written with s^2 in place of r. The remaining three equations become, with $r = s^2$,

$$u^3 a_3' = 1, \qquad u^4 a_4' = -s + 3r^2 = s + s^4, \qquad u^6 a_6' = r^3 - t - t^2 = s^6 + t + t^2.$$

So $a_3' = 1$ gives $u^3 = 1$; $a_4' = 0$ gives $s^4 = s$, i.e. $s \in \mathbb{F}_4$; and $a_6' = 0$ gives $t^2 + t = s^6$. Counting: if $s = 0$ then $t^2 + t = 0$, so $t \in \mathbb{F}_2$, two values. If $s \neq 0$ then $s \in \mathbb{F}_4^\times$ has $s^3 = 1$, so $s^6 = 1$ and $t^2 + t = 1$, whose two roots generate $\mathbb{F}_4$ over $\mathbb{F}_2$. Again two values. So there are $4 \cdot 2 = 8$ pairs (s, t) and 3 cube roots of unity u, giving $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$. All the parameters lie in $\mathbb{F}_4$, so all 24 are defined over $\mathbb{F}_4$.
Over $\mathbb{F}_2$ the count collapses: $u^3 = 1$ forces $u = 1$, and of $s \in \mathbb{F}_2$ only $s = 0$ survives, since $s = 1$ gives $s^6 = 1$ and $t^2 + t = 1$ has no root in $\mathbb{F}_2$. That leaves $s = 0$ with $t \in \mathbb{F}_2$, so 2 automorphisms — just ± 1, as for any curve.

Remark. From 2 to 24 on a quadratic extension: in characteristic 2 essentially none of the automorphism group is visible over the prime field.

(5) Write $\psi_{s,t} \coloneqq \Psi_{1,s^2,s,t}$ for the automorphisms with $u = 1$, so that $\psi_{s,t}(x, y) = (x + s^2, y + sx + t)$. Show that

$$\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s',\, t+t'+s's^2}, \qquad \psi_{s,t}^2 = \psi_{0,s^3},$$

and deduce that these 8 automorphisms form a copy of the quaternion group Q_8. For the remaining factor, let $\omega \in \mathbb{F}_4$ be a primitive cube root of unity and $\varphi_\omega \coloneqq \Psi_{\omega,0,0,0}$, which has order 3. Check that conjugation by it is

$$\varphi_\omega \circ \psi_{s,t} \circ \varphi_\omega^{-1} = \psi_{\omega s,\, t},$$

so φ_ω normalizes Q_8, fixes its center $\langle \psi_{0,1} \rangle$, and cycles the three pairs $\pm i, \pm j, \pm k$ — the order-3 automorphism of Q_8. Since $\langle \varphi_\omega \rangle \cap Q_8 = 1$, conclude that

$$\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) \cong Q_8 \rtimes \mathbb{Z}/3\mathbb{Z} \cong \mathrm{SL}_2(\mathbb{F}_3).$$

As before:

```
G2 := AutomorphismGroup(ChangeRing(E2, GF(4)));
IdentifyGroup(G2), GroupName(G2);        // <24, 3>  SL(2,3)
{* Order(g) : g in G2 *};                 // {* 1, 2, 3^^8, 4^^6, 6^^8 *}
```

Exactly one element of order 2 again. Explain why that alone pins down Q_8 among the groups of order 8, and why it does *not* by itself pin down $\mathrm{SL}_2(\mathbb{F}_3)$ among the groups of order 24 — what extra fact did you use?

SOLUTION. **$\mathrm{SL}_2(\mathbb{F}_3)$. The extra fact is a normal Q_8 on which $\mathbb{Z}/3\mathbb{Z}$ acts faithfully.** With $u = 1$ and $r = s^2$ the substitution (3.1) reads $\psi_{s,t}(x, y) = (x + s^2,\ y + sx + t)$. Composing,

$$\psi_{s',t'}\big(\psi_{s,t}(x,y)\big) = \big(x + s^2 + s'^2,\ y + sx + t + s'(x + s^2) + t'\big) = \big(x + (s+s')^2,\ y + (s+s')x + t + t' + s's^2\big),$$

using $s^2 + s'^2 = (s + s')^2$ in characteristic 2. That is $\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s',\, t+t'+s's^2}$. Squaring, $\psi_{s,t}^2 = \psi_{0,\, 2t+s^3} = \psi_{0,s^3}$. Now $\psi_{0,t}$ is $(x, y) \mapsto (x, y + t)$ with $t \in \mathbb{F}_2$, so $\psi_{0,0} = \mathrm{id}$ and $\psi_{0,1}$ is an involution. For $s \neq 0$ we have $s^3 = 1$, so $\psi_{s,t}^2 = \psi_{0,1} \neq \mathrm{id}$ and $\psi_{s,t}$ has order 4. So the group

of order 8 has exactly one element of order 2 and six of order 4, which characterizes Q_8 among groups of order 8. For the conjugation, φ_ω^{-1} is $(x,y) \mapsto (\omega x, y)$ because $\omega^{-2} = \omega$, so

$$\varphi_\omega\big(\psi_{s,t}(\omega x, y)\big) = \varphi_\omega(\omega x + s^2,\ y + s\omega x + t) = \big(x + (\omega s)^2,\ y + (\omega s)x + t\big) = \psi_{\omega s,\,t}(x,y),$$

using $\omega^3 = 1$ and $\omega^2 s^2 = (\omega s)^2$. So conjugation by φ_ω multiplies s by ω: it fixes $\psi_{0,0}$ and $\psi_{0,1}$, the center of Q_8, and cycles the three pairs $\pm i, \pm j, \pm k$, which is the order-3 automorphism of Q_8. As $\langle \varphi_\omega \rangle \cap Q_8 = 1$, we get $\mathrm{Aut}(E_{\overline{\mathbb{F}}_2}) = Q_8 \rtimes \mathbb{Z}/3\mathbb{Z}$, a group of order 24 with a normal Q_8 on which $\mathbb{Z}/3\mathbb{Z}$ acts faithfully. That is $\mathrm{SL}_2(\mathbb{F}_3)$, the binary tetrahedral group. `Magma`'s label `<24, 3>` is $\mathrm{SL}_2(\mathbb{F}_3)$, as computed by hand. As to the question in the part: among groups of order 8 a unique involution characterizes Q_8, but among groups of order 24 it does not — the dicyclic group of order 24 also has one — and what settled $\mathrm{SL}_2(\mathbb{F}_3)$ was the normal Q_8 with $\mathbb{Z}/3\mathbb{Z}$ acting faithfully on it.

Remark 3.6 (Both groups are unit groups)**.** Automorphisms are the units of the endomorphism ring, $\mathrm{Aut}(E_{\overline{k}}) = \mathrm{End}(E_{\overline{k}})^\times$. In characteristic 2 and 3 the curve with $j = 0$ has $\mathrm{End}(E_{\overline{k}})$ a maximal order in the quaternion algebra over $\mathbb{Q}$ that is not split exactly at p and ∞ [Sil09, Theorem V.3.1], [Voi21, Theorem 42.1.9]. For $p = 3$ that order is the one computed in Problem 2.9(5), and its unit group is the dicyclic group of order 12 of Problem 3.5(3). For $p = 2$ it is the Hurwitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\frac{1+i+j+ij}{2}$ in $(-1,-1 \mid \mathbb{Q})$ [Voi21, §11.1], whose unit group is the binary tetrahedral group of order 24 of Problem 3.5(5). This explains the shape of Theorem 3.2: when $\mathrm{End}(E_{\overline{k}})$ is commutative it is $\mathbb{Z}$ or an imaginary quadratic order, whose unit groups are the cyclic groups of orders 2, 4 and 6, giving its first three lines, while a non-commutative $\mathrm{End}(E_{\overline{k}})$ is a quaternion order, and its unit group need not be, and here is not, cyclic.

Both counts can be recovered from the orders alone, with no elliptic curve anywhere in the computation:

```
for d in [-1, -3] do
  Dd := QuaternionAlgebra<Rationals() | -1, d>;
  Od := MaximalOrder(Dd);
  print RamifiedPrimes(Dd), #UnitGroup(Od);
end for;
// [ 2 ] 24      <- Hurwitz order, characteristic 2
// [ 3 ] 12      <- the endomorphism ring of Problem 2.9, characteristic 3
```

So the 24 and the 12 of Theorem 3.2 are the unit counts of the two maximal quaternion orders.

Looking ahead. Curves whose Frobenius has $a = 0$, like the one of section 2, have a name and a theory of their own. They come later in the course, with Hasse's bound and the structure of End over $\mathbb{F}_q$. In characteristic 0 the third case of the classification never occurs, and $\mathrm{End}(E_{\overline{k}}) \neq \mathbb{Z}$ becomes complex multiplication. The oddness of $\rho_{E,\ell}$ from Problem 1.5(4) is the first hint of modularity.

References

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[Voi21] John Voight, *Quaternion algebras*, Graduate Texts in Mathematics, vol. 288, Springer, Cham, 2021. MR4279905 ↑